

Creamy Academy: Complete Platform Blueprint & Architecture

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Project: Creamy Academy — Premium Latte Art Learning Platform

Executive Summary

Creamy Academy is a **premium online learning platform** dedicated to teaching professional barista latte art techniques. The platform combines a branded, café-inspired user interface with a robust content management system, gated video delivery, and comprehensive student progress tracking. The architecture is built on a modern **React + Express + tRPC + MySQL** stack with OAuth authentication, ensuring a seamless and secure learning experience.

1. Platform Overview

1.1 Core Value Proposition

Creamy Academy delivers **structured, video-based latte art education** with the following key features:

- Gated Video Content:** All video lessons are accessible exclusively to authenticated users, ensuring content protection and exclusivity.
- Structured Learning Paths:** Courses are organized into logical modules (Milk Steaming, Heart, Tulip, Rosetta, Advanced Patterns) with progressive difficulty.
- Progress Tracking:** Students see real-time progress across courses, lessons, and modules, with a "Continue Learning" dashboard.
- Supplemental Resources:** Downloadable PDFs (guides, checklists) attached to individual lessons for offline reference.
- Admin Content Management:** Instructors can upload videos, manage course structures, and monitor student enrollment.

1.2 Target Users

User Type	Role	Permissions
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Public Visitor	Anonymous	View landing page, course previews, sign-up call-to-action
Enrolled Student	Authenticated User	Access all video lessons, download resources, track progress
Instructor/Admin	Admin User	Upload videos, manage courses, view student analytics

2. Visual Design & Branding

2.1 Color Palette

The platform uses a warm, premium barista-school aesthetic with the following color scheme:

Color	Hex Code	Usage
Coffee Brown	#6B4423	Primary brand color, headers, buttons
Latte Beige	#F5E6D3	Light backgrounds, cards, accents
Golden Crema	#D4A574	Highlights, hover states, secondary buttons
Dark Brown	#3D2817	Text, deep accents
Cream White	#FFFBF7	Main background, text backgrounds

2.2 Logo & Branding

The **Creamy Academy** logo features:

- A stylized stainless steel milk pitcher pouring a latte art heart/rosetta pattern into a coffee cup
- Premium serif typography with "CREAMY ACADEMY" as the primary text

- Tagline: "Latte Art Mastery"
- Transparent background for flexible placement across the platform

Logo Usage:

- Header navigation (compact version, 48px height)
- Landing page hero section (full version, 200px+ height)
- Email signatures and external communications

2.3 Typography

Element	Font	Size	Weight
Headings (H1–H3)	Playfair Display (Serif)	32–56px	700
Body Text	Inter (Sans-serif)	14–16px	400
Buttons & CTA	Inter	14–16px	600
Course Titles	Playfair Display	24–28px	700

3. Database Schema & Data Models

3.1 Core Tables

users

Stores user account information and authentication state.

SQL

```
CREATE TABLE users (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  openId VARCHAR(64) UNIQUE NOT NULL,  
  name TEXT,  
  email VARCHAR(320),  
  loginMethod VARCHAR(64),  
  role ENUM('user', 'admin') DEFAULT 'user',  
  createdAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  updatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
```

```
lastSignedIn TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

courses

Represents a complete course (e.g., "Latte Art Fundamentals").

SQL

```
CREATE TABLE courses (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  title VARCHAR(255) NOT NULL,  
  slug VARCHAR(255) UNIQUE NOT NULL,  
  description TEXT,  
  instructor VARCHAR(255),  
  instructorBio TEXT,  
  thumbnailUrl VARCHAR(500),  
  duration INT COMMENT 'Total duration in minutes',  
  level ENUM('beginner', 'intermediate', 'advanced') DEFAULT 'beginner',  
  createdAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  updatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP  
);
```

modules

Logical groupings within a course (e.g., "Milk Steaming", "Heart Patterns").

SQL

```
CREATE TABLE modules (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  courseId INT NOT NULL,  
  title VARCHAR(255) NOT NULL,  
  description TEXT,  
  displayOrder INT DEFAULT 0,  
  createdAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  FOREIGN KEY (courseId) REFERENCES courses(id) ON DELETE CASCADE  
);
```

lessons

Individual video lessons within a module.

SQL

```
CREATE TABLE lessons (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  moduleId INT NOT NULL,  
  title VARCHAR(255) NOT NULL,  
  description TEXT,  
  videoUrl VARCHAR(500),  
  duration INT COMMENT 'Duration in seconds',  
  displayOrder INT DEFAULT 0,  
  createdAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  updatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  FOREIGN KEY (moduleId) REFERENCES modules(id) ON DELETE CASCADE  
);
```

resources

Downloadable supplemental materials (PDFs, guides) attached to lessons.

SQL

```
CREATE TABLE resources (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  lessonId INT NOT NULL,  
  title VARCHAR(255) NOT NULL,  
  description TEXT,  
  fileUrl VARCHAR(500) NOT NULL,  
  fileType VARCHAR(50) COMMENT 'e.g., pdf, docx',  
  createdAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  FOREIGN KEY (lessonId) REFERENCES lessons(id) ON DELETE CASCADE  
);
```

enrollments

Tracks which students are enrolled in which courses.

SQL

```
CREATE TABLE enrollments (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  userId INT NOT NULL,  
  courseId INT NOT NULL,  
  enrolledAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  completedAt TIMESTAMP NULL,  
  UNIQUE KEY (userId, courseId),  
  FOREIGN KEY (userId) REFERENCES users(id) ON DELETE CASCADE,
```

```
FOREIGN KEY (courseId) REFERENCES courses(id) ON DELETE CASCADE
);
```

progress

Tracks individual lesson completion per student.

SQL

```
CREATE TABLE progress (
  id INT PRIMARY KEY AUTO_INCREMENT,
  userId INT NOT NULL,
  lessonId INT NOT NULL,
  completedAt TIMESTAMP NULL,
  watchedDuration INT COMMENT 'Seconds watched',
  lastWatchedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
  UNIQUE KEY (userId, lessonId),
  FOREIGN KEY (userId) REFERENCES users(id) ON DELETE CASCADE,
  FOREIGN KEY (lessonId) REFERENCES lessons(id) ON DELETE CASCADE
);
```

3.2 Relationships & Cardinality

Plain Text

```
users (1) —> (many) enrollments —> (1) courses
users (1) —> (many) progress —> (1) lessons
courses (1) —> (many) modules —> (1) lessons
lessons (1) —> (many) resources
lessons (1) —> (many) progress
```

4. User Flows & Features

4.1 Public Landing Page

URL: /

Access: Public (no authentication required)

Components:

- **Hero Section:** High-quality background image of latte art, tagline "Master the Art of Latte," and prominent "Sign Up Now" CTA button

- **Course Preview Cards:** Display 2–3 featured courses with thumbnail, title, instructor name, and "Learn More" button
- **Value Proposition:** Section highlighting key benefits (structured learning, expert instructors, lifetime access)
- **Testimonials:** Student quotes and success stories
- **FAQ Section:** Common questions about enrollment, access, and course structure
- **Footer:** Links to privacy policy, terms of service, contact information

Design Notes:

- Use warm color palette (coffee brown, latte beige, golden crema)
- Premium, café-inspired aesthetic with subtle textures
- Responsive design for mobile, tablet, and desktop

4.2 Authentication Flow

Sign-Up & Login: Manus OAuth integration (already configured in template)

Flow:

1. User clicks "Sign Up" or "Login" button on landing page
2. Redirected to Manus OAuth portal
3. User authenticates (email/password or social login)
4. OAuth callback creates/updates user record in database
5. Session cookie set, user redirected to dashboard or course page

Protected Routes: All video player, dashboard, and admin pages require authentication

4.3 Course Catalog & Discovery

URL: `/courses`

Access: Authenticated users only

Components:

- **Course Grid:** Display all available courses with thumbnail, title, instructor, level badge, and "Enroll" button
- **Filters:** Filter by level (Beginner, Intermediate, Advanced)
- **Search Bar:** Full-text search across course titles and descriptions
- **Enrollment Status:** Show "Enrolled" badge for already-enrolled courses, "Continue Learning" button for in-progress courses

Data Fetched:

- List of all courses with metadata
- User's enrollment status for each course

4.4 Course Detail Page

URL: /courses/:courseId

Access: Authenticated users only

Components:

- **Header:** Course title, instructor name, instructor bio, course thumbnail
- **Course Info Section:** Description, learning outcomes, course level, total duration
- **Module & Lesson List:** Hierarchical display of all modules and lessons with duration, completion status
- **Enroll Button:** Prominent CTA if user is not enrolled; "Continue Learning" if enrolled
- **Instructor Card:** Photo, name, credentials, bio

Data Fetched:

- Course details (title, description, instructor info)
- All modules and lessons for the course
- User's enrollment status and progress

4.5 Gated Video Player Page

URL: /courses/:courseId/lessons/:lessonId

Access: Authenticated users enrolled in the course only

Components:

- **Video Player:** Custom HTML5 video player with:
 - Play/pause controls
 - Playback speed selector (0.5x, 1x, 1.5x, 2x)
 - Progress bar with seek functionality
 - Fullscreen toggle
 - Volume control
 - Captions (if available)
- **Lesson Sidebar:** List of all lessons in the current module with:
 - Completion status (checkmark for completed lessons)

- Current lesson highlight
- Click-to-navigate to other lessons
- **Lesson Details:** Title, description, duration, instructor
- **Resources Section:** Downloadable PDFs and supplemental materials for the lesson
- **Next Lesson Button:** Quick navigation to the next lesson in sequence

Authentication Guard:

- Verify user is authenticated
- Verify user is enrolled in the course
- If not, redirect to login or course detail page

Progress Tracking:

- Record lesson view timestamp
- Track watch duration (for analytics)
- Mark lesson as "completed" when user reaches 80%+ of video duration

4.6 Student Dashboard

URL: /dashboard

Access: Authenticated users only

Components:

- **Welcome Section:** Personalized greeting ("Welcome back, [Name]!")
- **Continue Learning:** Display in-progress courses with:
 - Course thumbnail
 - Current lesson title
 - Progress bar (% of course completed)
 - "Resume" button linking to the last watched lesson
- **Enrolled Courses:** Grid of all enrolled courses with:
 - Course thumbnail
 - Title and instructor
 - Progress percentage
 - "View Course" button
- **Learning Stats:** Summary cards showing:
 - Total courses enrolled

- Total lessons completed
- Total watch time (hours)
- Current streak (days of consecutive learning)

Data Fetched:

- User's enrollments
- Progress data for each enrolled course
- Last watched lesson for each course

4.7 Admin Panel

URL: /admin

Access: Admin users only (role = 'admin')

Components:

Dashboard Tab

- Overview statistics: Total students, total courses, total lessons, total watch hours
- Recent enrollments table
- Top-performing courses chart

Courses Tab

- List of all courses with actions: Edit, Delete, View Details
- "Create New Course" button
- Course form: Title, description, instructor, thumbnail upload, level

Modules Tab

- Select a course, then manage its modules
- Add/edit/delete modules
- Drag-to-reorder modules

Lessons Tab

- Select a course and module, then manage lessons
- Add/edit/delete lessons
- Video upload interface (with progress indicator)
- Lesson form: Title, description, video URL, duration

Resources Tab

- Select a lesson, then manage its downloadable resources
- Add/edit/delete resources
- PDF/file upload interface
- Resource form: Title, description, file upload

Students Tab

- List of all enrolled students with:
 - Name, email, enrollment date
 - Courses enrolled in
 - Total watch time
 - Last activity date
- Search and filter functionality

5. Technical Architecture

5.1 Technology Stack

Layer	Technology	Purpose
Frontend	React 19 + Tailwind CSS 4	UI components, responsive design
Backend	Express 4 + tRPC 11	API layer, type-safe RPC procedures
Database	MySQL (via Drizzle ORM)	Data persistence
Authentication	Manus OAuth	User identity and session management
File Storage	S3 (via Manus storage helpers)	Video and PDF hosting
Video Hosting	S3 (via Manus storage helpers)	Streaming video content

5.2 API Architecture (tRPC Procedures)

All backend logic is exposed via tRPC procedures under `/api/trpc`. Key procedure groups:

auth

- `auth.me` — Get current user info
- `auth.logout` — Clear session

courses

- `courses.list` — Get all courses (public)
- `courses.detail` — Get course with modules and lessons (authenticated)
- `courses.enroll` — Enroll user in a course (authenticated)
- `courses.create` — Create new course (admin only)
- `courses.update` — Update course details (admin only)
- `courses.delete` — Delete course (admin only)

modules

- `modules.list` — Get modules for a course (authenticated)
- `modules.create` — Create module (admin only)
- `modules.update` — Update module (admin only)
- `modules.delete` — Delete module (admin only)

lessons

- `lessons.list` — Get lessons for a module (authenticated)
- `lessons.detail` — Get single lesson with resources (authenticated)
- `lessons.create` — Create lesson (admin only)
- `lessons.update` — Update lesson (admin only)
- `lessons.delete` — Delete lesson (admin only)

resources

- `resources.list` — Get resources for a lesson (authenticated)
- `resources.create` — Upload resource file (admin only)
- `resources.delete` — Delete resource (admin only)

progress

- `progress.list` — Get user's progress across courses (authenticated)
- `progress.markComplete` — Mark lesson as completed (authenticated)
- `progress.getStats` — Get user's learning statistics (authenticated)

admin

- `admin.stats` — Get platform statistics (admin only)
- `admin.students` — Get list of all students (admin only)
- `admin.enrollments` — Get enrollment analytics (admin only)

5.3 Frontend Component Structure

Plain Text

```
client/src/
├── pages/
│   ├── Home.tsx           # Landing page
│   ├── Courses.tsx        # Course catalog
│   ├── CourseDetail.tsx   # Course detail & enrollment
│   ├── LessonPlayer.tsx   # Gated video player
│   ├── Dashboard.tsx      # Student dashboard
│   ├── Admin/
│   │   ├── AdminDashboard.tsx # Admin overview
│   │   ├── CourseManager.tsx  # Course CRUD
│   │   ├── LessonManager.tsx  # Lesson CRUD
│   │   └── StudentManager.tsx # Student analytics
│   └── NotFound.tsx
├── components/
│   ├── VideoPlayer.tsx     # Custom video player
│   ├── CourseCard.tsx      # Reusable course card
│   ├── LessonList.tsx      # Lesson sidebar
│   ├── ProgressBar.tsx     # Progress visualization
│   └── ResourceDownload.tsx # Resource download UI
├── hooks/
│   ├── useAuth.ts          # Auth state management
│   ├── useCourseProgress.ts # Progress tracking
│   └── useVideoPlayer.ts    # Video player logic
├── lib/
│   └── trpc.ts              # tRPC client setup
└── App.tsx                  # Routes & layout
```

5.4 File Storage Strategy

Videos:

- Upload via admin panel → stored in S3 via `storagePut()`
- Retrieve presigned URL via `storageGet()` for streaming
- Store video URL in `lessons.videoUrl` database field

Resources (PDFs):

- Upload via admin panel → stored in S3 via `storagePut()`
- Store file URL in `resources.fileUrl` database field
- Frontend provides direct download link

Course Thumbnails:

- Upload via admin panel → stored in S3
- Store thumbnail URL in `courses.thumbnailUrl` database field

6. Security & Access Control

6.1 Authentication

- **Method:** Manus OAuth (configured in template)
- **Session:** Secure HTTP-only cookie with CSRF protection
- **Token Expiry:** Configurable via Manus platform settings

6.2 Authorization

Route/Feature	Public	User	Admin
Landing page	✓	✓	✓
Course catalog	✗	✓	✓
Video player	✗	✓*	✓
Dashboard	✗	✓	✓
Admin panel	✗	✗	✓

*User must be enrolled in the course to access video player

6.3 Data Privacy

- User passwords: Managed by Manus OAuth (not stored locally)

- Progress data: Scoped to authenticated user only
 - Admin operations: Require `role = 'admin'` check in tRPC procedures
 - Video content: Served via S3 presigned URLs (time-limited access)
-

7. Deployment & Hosting

7.1 Deployment Platform

Creamy Academy is deployed on **Manus Hosting** with the following features:

- **Frontend:** Vite-built React app served via CDN
- **Backend:** Express server running on Node.js runtime
- **Database:** MySQL database (managed)
- **Storage:** S3 integration for videos and PDFs
- **SSL/TLS:** Automatic HTTPS with custom domain support
- **Auto-scaling:** Handles traffic spikes automatically

7.2 Environment Variables

Variable	Purpose	Example
<code>DATABASE_URL</code>	MySQL connection string	<code>mysql://user:pass@host/db</code>
<code>JWT_SECRET</code>	Session cookie signing secret	Auto-generated
<code>VITE_APP_ID</code>	Manus OAuth app ID	Auto-generated
<code>OAUTH_SERVER_URL</code>	OAuth backend URL	Auto-generated

8. Development Workflow

8.1 Local Development

Bash

```
# Install dependencies  
pnpm install
```

```
# Start dev server (frontend + backend)
pnpm dev

# Run tests
pnpm test

# Build for production
pnpm build
```

8.2 Database Migrations

1. Update schema in `drizzle/schema.ts`
2. Generate migration: `pnpm drizzle-kit generate`
3. Apply migration via Manus UI or CLI

8.3 Adding Features

1. Define database schema changes
2. Create tRPC procedures in `server/routers.ts`
3. Build React components in `client/src/pages/` or `client/src/components/`
4. Wire components to tRPC procedures via `trpc.*.useQuery/useMutation`
5. Test with Vitest

9. Future Enhancements

9.1 Phase 2 Features (Post-Launch)

- **Certificates:** Auto-generate completion certificates for finished courses
- **Community Forum:** Discussion board for students to share latte art photos and tips
- **Live Sessions:** Scheduled live Q&A with instructors
- **Quizzes & Assessments:** Knowledge checks at the end of each module
- **Mobile App:** Native iOS/Android apps for offline lesson viewing
- **Instructor Profiles:** Public instructor pages with bio, credentials, and social links

9.2 Monetization Options

- **Subscription Model:** Monthly/annual subscription for unlimited course access
- **Pay-Per-Course:** One-time purchase for individual courses

- **Corporate Training:** Bulk licensing for coffee shops and barista schools
- **Affiliate Program:** Commission for referrals

10. Success Metrics

Metric	Target	Measurement
Course Completion Rate	>70%	Lessons completed / lessons enrolled
Student Engagement	>5 hrs/month	Average watch time per student
Enrollment Growth	+50/month	New enrollments per month
Platform Uptime	99.9%	Downtime / total time
Video Load Time	<3 sec	Time to first frame

11. Conclusion

Creamy Academy is a **comprehensive, production-ready platform** for premium latte art education. The architecture prioritizes security, scalability, and user experience, with a clear roadmap for future enhancements. The warm, café-inspired design creates an inviting learning environment, while the robust backend ensures reliable content delivery and student progress tracking.

References

1. [Scoolinary Latte Art Course](#) — Reference for course structure and video delivery UX
2. [Typsy Latte Art Fundamentals](#) — Reference for lesson organization and progress tracking
3. [Manus Web App Template Documentation](#) — Technical stack and authentication reference